

120 Days of Quantitative Finance

Month 4: Stochastic Calculus & Time Series Modelling

Week 13: Stochastic Processes & Brownian Motion

Day 87: Brownian Motion (Wiener Process)

1 Introduction

Brownian Motion, also known as the Wiener Process, is the fundamental building block of continuous-time stochastic models.

It is used extensively in financial mathematics to model the random behavior of asset prices.

2 Definition

A stochastic process $\{W_t\}_{t \geq 0}$ is called a Brownian Motion if it satisfies:

1. $W_0 = 0$
2. Independent increments:

$$W_t - W_s \perp \mathcal{F}_s \quad \text{for } t > s$$

3. Normally distributed increments:

$$W_t - W_s \sim \mathcal{N}(0, t - s)$$

4. Continuous paths

3 Key Properties

3.1 Distribution

$$W_t \sim \mathcal{N}(0, t)$$

Mean:

$$E[W_t] = 0$$

Variance:

$$Var(W_t) = t$$

3.2 Scaling Property

$$W_{ct} \sim \sqrt{c}W_t$$

This shows how variance scales with time.

3.3 Stationary Increments

$$W_{t+h} - W_t \sim \mathcal{N}(0, h)$$

3.4 Non-Differentiability

Brownian Motion is continuous but nowhere differentiable.

This is a crucial property that leads to Ito calculus.

4 Discrete Approximation

Brownian Motion can be approximated as:

$$W_{t+\Delta t} = W_t + \sqrt{\Delta t} \cdot \epsilon_t$$

where:

$$\epsilon_t \sim \mathcal{N}(0, 1)$$

This links discrete random walks to continuous-time processes.

5 Quadratic Variation

One of the most important properties:

$$\sum (W_{t_{i+1}} - W_{t_i})^2 \rightarrow t$$

This result is fundamental for stochastic calculus.

6 Intuition

Brownian Motion represents:

- continuous random movement
- no memory (Markov property)
- increasing uncertainty over time

It can be visualized as a particle undergoing random motion.

7 Application in Finance

Brownian Motion is used to model:

- stock price movements
- interest rates
- volatility dynamics

It forms the basis of:

- Geometric Brownian Motion (GBM)
- Black-Scholes model
- Stochastic Differential Equations

8 Conclusion

Brownian Motion is the cornerstone of continuous-time finance.

Understanding its properties is essential before moving to:

- Ito's Integral
- Ito's Lemma
- Stochastic Differential Equations